



Let X be a $K3$ surface such that the lattice S_X is of rank 11 with the Gram matrix

$$\text{diag}(2, -2, -2, \dots, -2),$$

and the period $\omega_X \in T_X \otimes \mathbb{C}$ is general so that

$$\text{O}(T_X, \omega_X) = \{\pm \text{id}\}.$$

The lattice S_X is isomorphic to $I_{1,10}(2)$. Let $e_0, e_1, \dots, e_{10}$ be the vectors of $I_{1,10}$ such that

$$\langle e_0, e_0 \rangle = 1, \quad \langle e_i, e_i \rangle = -1 \quad (i = 1, \dots, 10), \quad \langle e_i, e_j \rangle = 0 \quad (i \neq j).$$

By Vinberg algorithm, we see that $I_{1,10}$ is generated by the vectors

$$\begin{aligned} \alpha_0 &= e_0 - e_1 - e_2 - e_3, \\ \alpha_1 &= e_1 - e_2, \\ &\dots \\ \alpha_9 &= e_9 - e_{10}, \\ \alpha_{10} &= e_{10}, \\ \alpha_{11} &= 3e_0 - e_1 - \dots - e_{10}, \end{aligned}$$

of norms $-2, -2, \dots, -2, -1, -1$, and $\text{O}(I_{1,10}, \mathcal{P})$ is generated by the reflections with respect to these vectors. Hence $\text{O}(S_X, \mathcal{P})$ is also generated by these 12 reflections. The order of $\text{O}(q_S)$ is

$$D := 46998591897600.$$

The subgroup of $\text{O}(q_S)$ generated by the image of these 12 reflections by the natural homomorphism $\eta_S: \text{O}(S_X, \mathcal{P}) \rightarrow \text{O}(q_S)$ is also equal to D . Hence η_S is surjective, and the kernel of η_S is of index D in $\text{O}(S_X, \mathcal{P})$. Note that the discriminant group of S_X is 2-elementary, and hence the period condition for $g \in \text{O}(S_X, \mathcal{P})$ is $\eta_S(g) = \text{id}$. Therefore we see that the image of

$$\text{Aut}(X) \rightarrow \text{O}(S_X, \mathcal{P})$$

is equal to the subgroup of the kernel of η_S consisting of elements that preserve the nef-and-big cone N_X .

No.	<i>ADE</i>	$O(R)$	$i(\eta_R)$	m_2	m_4
1	$A_1 + D_6$	64210599936000	67456	62	612
2	A_5	133772083200	505920	30	804
3	$A_1 + A_3 + D_4$	321052999680	2698240	38	852
4	$3A_1 + D_5$	535088332800	2698240	46	1028
5	$3A_1 + D_4$	31708938240	4553280	30	932
6	$3A_1$	127401984	5902400	6	1044
7	$A_1 + A_3$	707788800	6374592	14	964
8	$11A_1 + D_4$	2853804441600	6475776	46	772
9	$A_1 + A_4$	3483648000	6475776	22	1012
10	$A_2 + A_3$	2090188800	6475776	18	924
11	$5A_1 + A_3$	9555148800	7555072	22	884
12	$7A_1$	743178240	16189440	14	964
13	$2A_1 + A_2$	99532800	22665216	10	1004
14	$A_1 + 2A_2$	298598400	22665216	14	1092
15	$15A_1$	67947724800	22665216	30	932
16	$7A_1$	452984832	26560800	14	1092
17	$11A_1 + D_4$	570760888320	32378880	46	1028
18	$4A_1 + A_5$	28665446400	37775360	38	1236
19	$5A_1 + A_3$	1698693120	42497280	22	1012
20	$9A_1 + D_6$	8561413324800	64757760	78	1476
21	$3A_1 + 2A_3$	5096079360	84994560	30	1188
22	$15A_1$	16307453952	94438400	30	1188
23	$5A_1 + A_3$	743178240	97136640	22	1140
24	$7A_1 + 2D_4$	1712282664960	129515520	62	1380
25	$11A_1 + D_4$	67947724800	271982592	46	1284

TABLE 0.1. Genus of R

The genus of the orthogonal complement $R = (S_X \hookrightarrow L_{26})^\perp$ of a primitive embedding $\iota: S_X \hookrightarrow L_{26}$ does not depend on the choice of the embedding ι , and the mass of this genus is

$$\sum_R \frac{1}{|O(R)|} = \frac{3773551}{128421199872000},$$

where the summation is taken over a representative set of isomorphism classes in the genus. By Kneser's neighboring method, we enumerate all the isomorphism classes of lattices in this genus. The result is given in Table 0.1. Here m_ν is the size of $R_\nu := \{v \in R \mid \langle v, v \rangle = -\nu\}$ for $\nu = 2, 4$, and *ADE* is the *ADE*-type of the set of roots R_2 . The index $i(\eta_R)$ of the image of the natural homomorphism $\eta_R: O(R) \rightarrow O(q_R)$ is also presented.

Remark 0.1. For the calculation of the sizes of $O(R)$, we used **MAGMA**. This is my first usage of **MAGMA** (the web calculator).

We say that two primitive embeddings $\iota_1: S_X \hookrightarrow L_{26}$ and $\iota_2: S_X \hookrightarrow L_{26}$ are equivalent if there exist isometries $g_S: S_X \cong S_X$ and $g_L: L_{26} \cong L_{26}$ such that $\iota_1 g_L = g_S \iota_2$. (Note that we write mappings from the right.) Suppose that we fix a lattice R in the above genus. Then q_S and $-q_R$ are isomorphic. Since the natural homomorphism $O(S, \mathcal{P}) \rightarrow O(q_S)$ is surjective, we see that the action of $O(S, \mathcal{P})$ on the set of isomorphisms $q_S \cong -q_R$ is transitive. Hence the equivalence class of primitive embeddings $S_X \hookrightarrow L_{26}$ is determined solely by the isomorphism class of the orthogonal complement $R = (S_X \hookrightarrow L_{26})^\perp$.